Introductory Statistics MID

Show equations, substitutions, intermediate reasoning, and a final labeled answer.

1. [8 pts] For observations [2, 2, 3, 10, 8], compute the sample mean and sample variance. Show deviations or an equivalent calculation.

2. [7 pts] A batch has 5 defective and 11 good units. Two are sampled without replacement. Compute P(both defective) and explain why the second denominator changes.

3. [8 pts] A sample of n=40 has p■=0.55. Construct a 95% normal-approximation interval and state what the interval means.

4. [6 pts] A confidence interval is wide even though the point estimate is stable. Give two concrete changes that would narrow it and one trade-off.

5. [6 pts] A p-value is 0.03 for a pre-registered test at α=0.05. State the decision and the conclusion without claiming that the null is proven false.

6. [8 pts] For observations [7, 9, 6, 9, 4], compute the sample mean and sample variance. Show deviations or an equivalent calculation.

7. [7 pts] A batch has 5 defective and 14 good units. Two are sampled without replacement. Compute P(both defective) and explain why the second denominator changes.

8. [8 pts] A sample of n=40 has p■=0.35. Construct a 95% normal-approximation interval and state what the interval means.